

One of the most important things about *Mathematica* is that it is highly extensible. As well as using the functions that are already built into *Mathematica*, you can define your own functions, to customize *Mathematica* to your particular needs.

This Appendix contains examples from packages that give various kinds of *Mathematica* definitions. Most of the examples are just fragments of packages, but are nevertheless chosen to be representative of the style used in the whole packages.

The examples are “real”, in the sense that they come from actual *Mathematica* packages distributed with many versions of *Mathematica*.

The topics for the packages in this Appendix have been chosen to be of comparatively widespread applicability. Many actual *Mathematica* packages will address much more specific application areas. One important point about the packages in this Appendix is that they are all plain *Mathematica* input files, not “notebooks”. Notebooks can include not only *Mathematica* definitions, but also explanatory text and other material, as discussed briefly in Section 1.3. The details of writing notebooks for *Mathematica* on various computers will be described elsewhere.

About the illustration overleaf:

This is a parametric plot of the spherical harmonic $Y_3^1(\theta, \phi)$. The spherical harmonic is represented in *Mathematica* by the function `SphericalHarmonicY[3, 1, theta, phi]`. With this function assigned as the value of `y`, the plot was generated by the command

`ParametricPlot3D[list, thetarange, phirange]`, where `list` is `{Abs[y] Sin[theta] Cos[phi], Abs[y] Sin[theta] Sin[phi], Abs[y] Cos[theta]}`, `thetarange` is `{theta, Pi/30, Pi-Pi/30, Pi/30}` and `phirange` is `{phi, 0, 2Pi, Pi/15}`. The function `ParametricPlot3D` is defined in the package `ParametricPlot.m`.